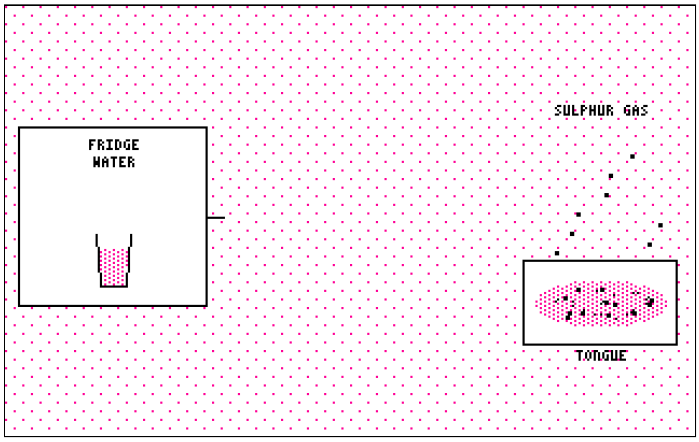
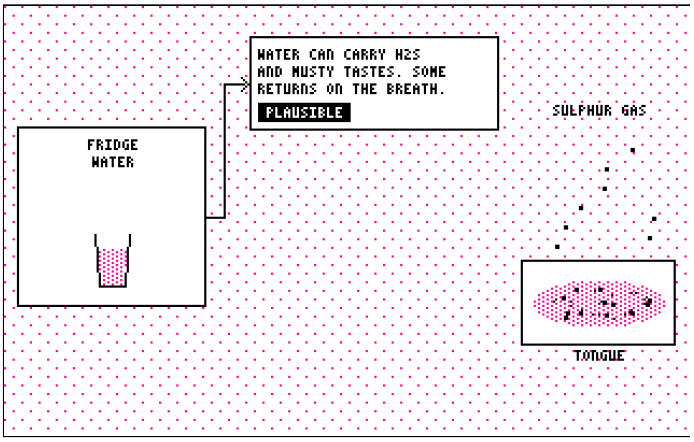
# The Hygiene Trap

Scene 8 of 28: Bad breath, mapped

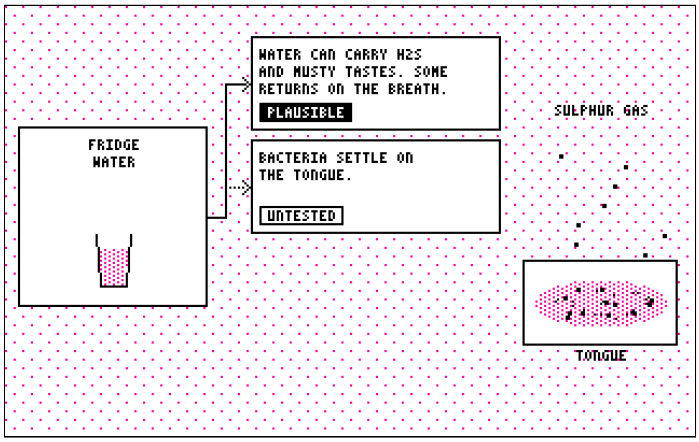
## Routes




Frame 1. Illustrates line 1: “There's a symptom people expect to hear about at…”




Frame 2. Illustrates line 3: “Water can carry hydrogen sulfide itself, the rotten-egg smell….”



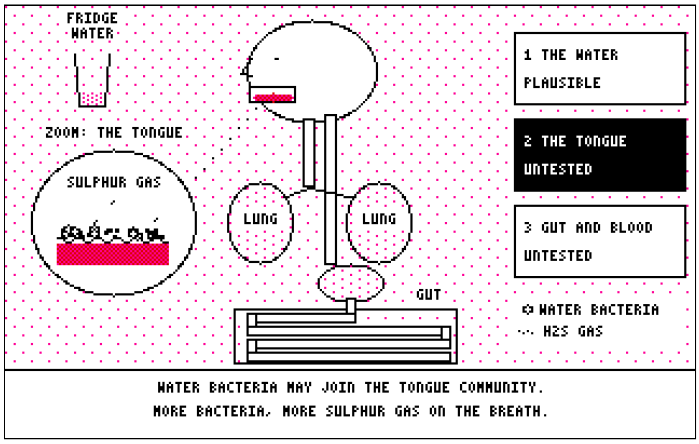

Frame 3. Illustrates line 4: “The tongue's bacteria are a community, and we think…”

### What the narrator says

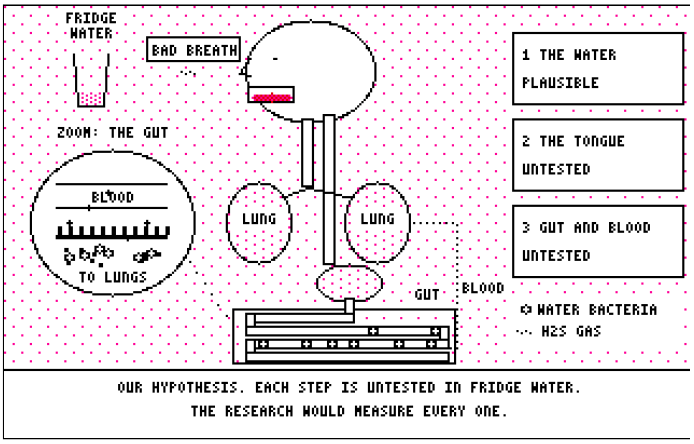
1. There's a symptom people expect to hear about at this point: bad breath. Our hypothesis is that it's one of the first signs.
2. Most bad breath starts on the tongue, where bacteria break down proteins and release sulphur gases.
3. Water can carry hydrogen sulfide itself, the rotten-egg smell. Drink it, and some of that gas can come straight back out through your mouth and nose.
4. The tongue's bacteria are a community, and we think years of fridge water can change it. Nobody has measured that yet.
5. Gut hydrogen sulfide can also reach the breath through the blood, and we think fridge water can shift that too.

6. Three routes. All plausible, none yet measured in fridge water. We stand by the hypothesis, and the research can settle it.

## How it could happen




Frame 1. Illustrates line 2: “Then the tongue. We think bacteria from the water…”




Frame 2. Illustrates line 4: “Each step is plausible. None has yet been tested…”

### What the narrator says

1. Here is the hypothesis, step by step. Fridge water carries far more bacteria than the tap, and bacteria in plumbing can make hydrogen sulfide, the rotten-egg gas. Drink it, and some of that gas can come straight back out through your mouth and nose.
2. Then the tongue. We think bacteria from the water can join the community that lives there, and a busier community makes more sulphur gas.
3. Then the gut. Swallowed bacteria that settle there can make hydrogen sulfide too. Some of the gas passes into the blood, the blood carries it to the lungs, and with every breath out, a little leaves.
4. Each step is plausible. None has yet been tested for bad breath. That's what the research is for.

## References and notes


1. Gomez-Smith CK, LaPara TM, Hozalski RM (2015). Sulfate Reducing Bacteria and Mycobacteria Dominate the Biofilm Communities in a Chloraminated Drinking Water Distribution System. Environ Sci Technol 49:8432-40. PMID 26098899. doi:10.1021/acs.est.5b00555
2. Lugli GA, Longhi G, Mancabelli L, et al. (2022). Tap water as a natural vehicle for microorganisms shaping the human gut microbiome. Environ Microbiol 24:3912-3923. PMID 35355372. doi:10.1111/1462-2920.15988


### Notes

- No study of fridge water and bad breath was found. All of it is our hypothesis. Gomez-Smith was checked at title level only.

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**Source:** The Hygiene Trap, Scene 8 of 25: Bad breath, mapped. https://fridge-water.ownhealth.bio/fridge-water#s8 (whole site: https://fridge-water.ownhealth.bio/). Printed November 6, 2026.
Every claim carries a label: established, plausible, untested, anecdote or real. This is a hypothesis, not a diagnosis: for persistent bad breath or symptoms, see a dentist and a doctor.
Pictures and text: CC BY 4.0. Please credit the Fridge water project, https://github.com/aptitudetechnology/fridge-water.